

From Green Tea to Targeted Therapy: Integrating Epigallocatechin Gallate (EGCG) into Modern Oncology

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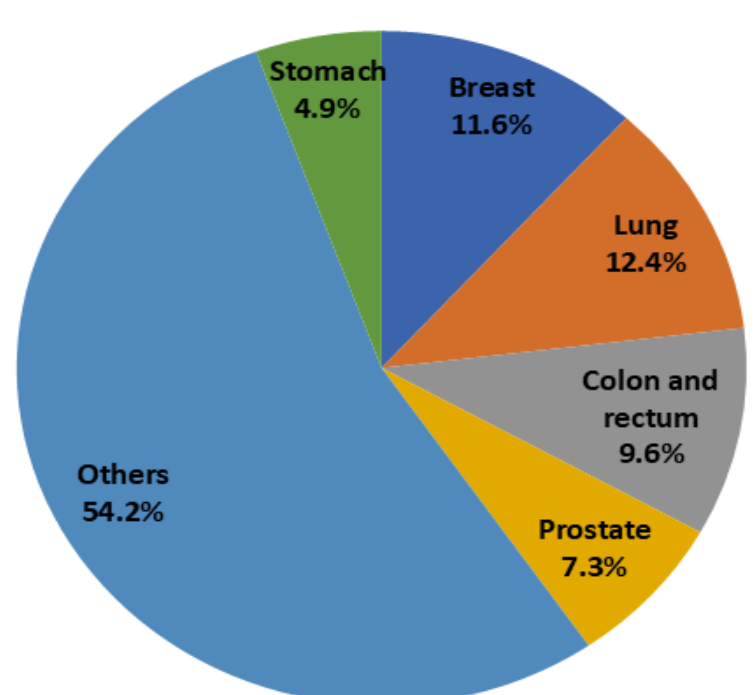
AIM OF THE STUDY

The current poster reviews published studies highlighting the anticancer potential of Epigallocatechin Gallate (EGCG) from green tea, with a focus on its role in cancer prevention, basic molecular mechanisms, and its possible use as a supportive agent in modern cancer therapy

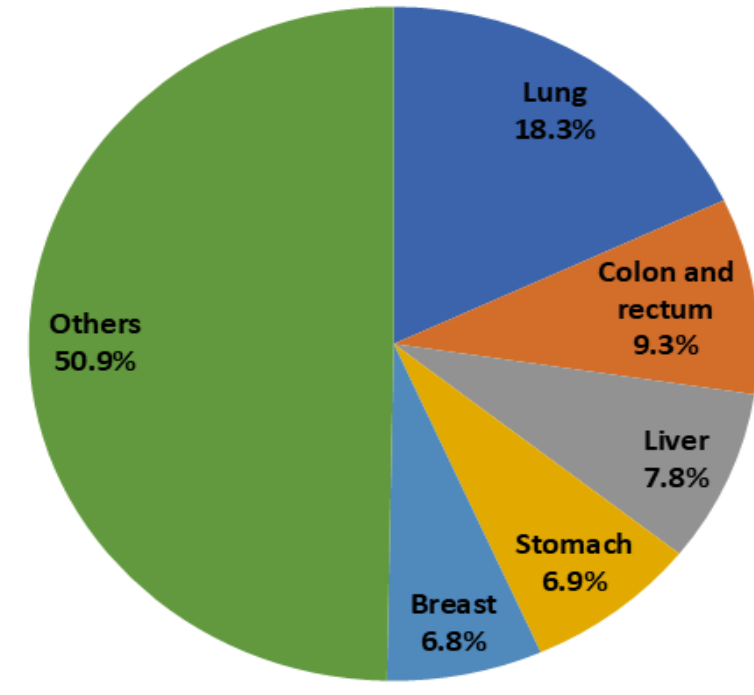
INTRODUCTION

- Cancer remains one of the leading causes of morbidity and mortality worldwide, prompting continuous exploration of novel, safer, and more effective therapeutic and preventive strategies alongside conventional chemotherapy and radiotherapy
- Because of their multitargeted activities, reduced safety profiles, and potential to improve the effectiveness of already available anticancer medications, naturally derived phytochemicals have garnered substantial interest in oncology research in recent decades
- Due in large part to its high concentration of polyphenolic chemicals called catechins, green tea (*Camellia sinensis*) has been drunk for centuries and is widely known for its health-promoting qualities
- The most prevalent and physiologically active of these catechins, epigallocatechin-3-gallate (EGCG), is responsible for much of the antioxidant and medicinal properties of green tea
- EGCG has potent anti-inflammatory, metal-chelating, and antioxidant qualities that allow it to regulate cellular redox balance and guard against oxidative stress and DNA damage, all of which are strongly associated with carcinogenesis
- By controlling important molecular pathways involved in cell proliferation, apoptosis, angiogenesis, metastasis, and immunological regulation, EGCG can prevent cancer initiation, development, and progression, according to numerous in vitro, in vivo, and epidemiological studies
- More research into EGCG's mechanisms of action, therapeutic uses, and clinical prospects is necessary because of the mounting body of evidence that shows it to be a viable adjuvant and preventative agent in cancer oncology

Most common cancers in the world



Most common causes of cancer death in the world



GLOBOCAN 2022: New Global Cancer Data

METHODOLGY

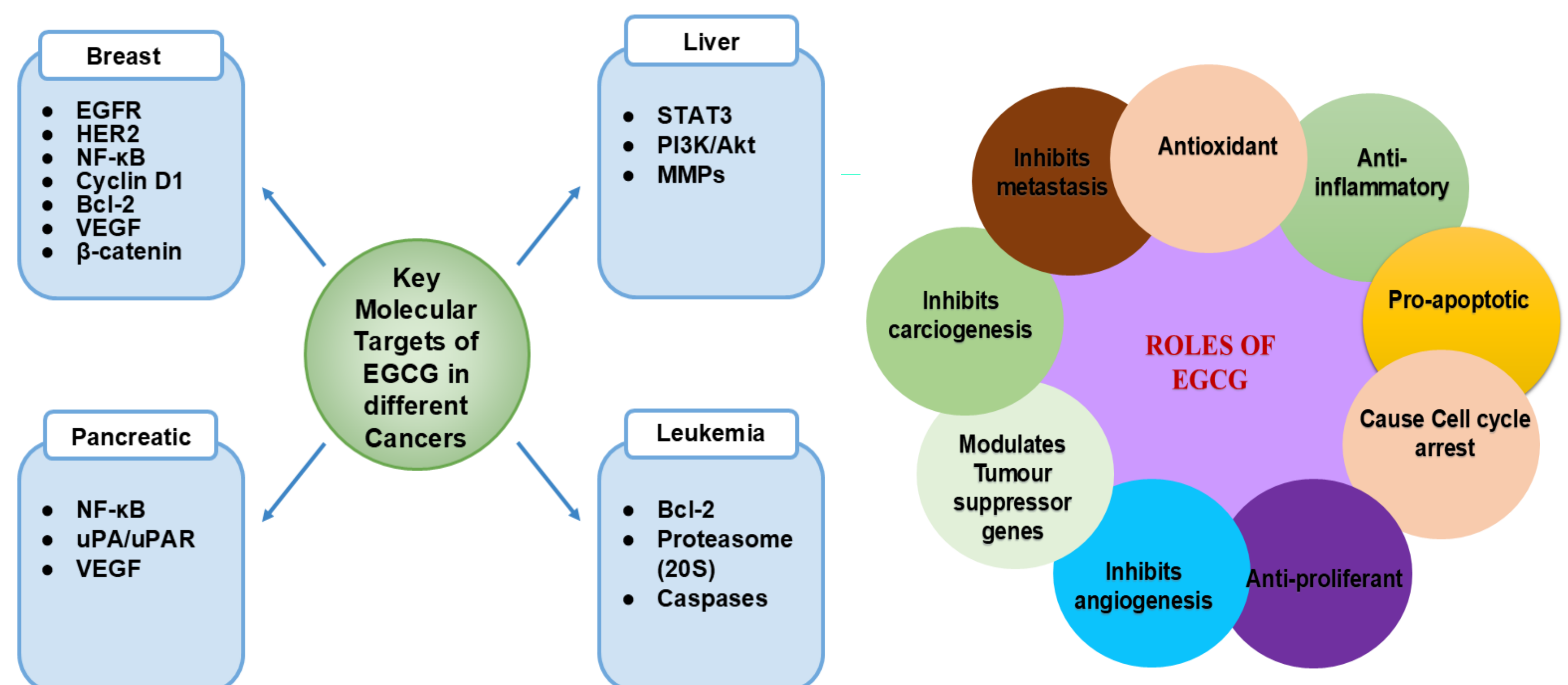
- Literature survey was done using PubMed and Google Scholar
- The papers that are selected for reviewing includes the studies related to cancer and EGCG
- Studies published during the last ten years have been more emphasized
- Papers including literature review, systematic review, research papers, meta-analysis, comparative and cohort studies, limited to English language were referred

Effects of EGCG on Different Cancer Types

Type of Cell	Concentration used	Targeted hallmark	Outcome
Breast cancer	30 μ mol/L	Apoptosis	EGCG suppressed the proliferation of human MCF-7 breast cancer cells
Gastric cancer	80 μ g/mL	Apoptosis	Induced apoptosis of gastric cancer cells via down-regulating HIF-1 α
Nasopharyngeal cancer	40 μ M	Apoptosis	Inhibited the growth of nasopharyngeal cancer cells through the inhibition of the SIRT1-p53 signaling pathway
Pancreatic cancer	40 μ g/mL	Apoptosis	Inhibited proliferation and induced apoptosis in pancreatic cancer cells via PTEN
Hepatocellular carcinoma	0.5 mM	Apoptosis	Induced an inhibitory effect on activity of the de novo lipogenesis (DNL) pathway, mediating apoptosis in HepG2 cells
Endometrial cancer	60 μ M	Angiogenesis inhibition	Reduced the secretion of VEGFA from cancerous cells

RESULTS & DISCUSSION

- EGCG's role as a multitargeted phytochemical in oncology research is supported by several in vitro studies using a variety of cancer cell lines, which show that it has strong anticancer effects in a dose- and time-dependent manner
- The majority of research indicates that EGCG has biologically significant effects between 10 and 100 μ M. Higher concentrations (50–100 μ M) cause significant apoptosis and growth inhibition, but lower quantities (10–25 μ M) mainly show cytostatic effects. For instance, lung cancer (A549) and colorectal cancer (HT-29, HCT116) cells frequently need 40–80 μ M EGCG for substantial apoptotic responses, while breast cancer cell lines including MCF-7 and MDA-MB-231 exhibit decreased proliferation at ~20–40 μ M
- According to mechanistic study, EGCG increases apoptotic signals and decreases cell viability by downregulating the PI3K/Akt/mTOR pathway. Prostate (PC-3) and lung (A549) cancer cell lines treated with $\geq$ 40 μ M EGCG consistently show suppression of Akt phosphorylation
- The intrinsic mitochondrial apoptotic pathway, which is marked by elevated Bax/Bcl-2 ratio, cytochrome c release, and activation of caspase-3 and -9, is triggered by EGCG administration. At doses greater than 50 μ M, these effects are visible in liver (HepG2) and cervical (HeLa) carcinoma cells
- EGCG also inhibits NF- κ B activation and disrupts MAPK/ERK signaling, which lowers the expression of VEGF, MMPs, and cyclin D1. This leads to a reduction in angiogenic potential, invasion, and cell cycle progression
- Taken together, these findings demonstrate EGCG's capacity to concurrently target several carcinogenic pathways at micromolar concentrations that are feasible in experimental systems. The consistent suppression of proliferation and induction of apoptosis highlight EGCG's potential as an adjuvant therapeutic drug and serve as a basis for next translational and combinatorial research in cancer oncology, even though effective doses vary among various cell types



CONCLUSIONS

- ✓ Epigallocatechin-3-gallate (EGCG), a significant bioactive component of green tea, has strong anticancer activity against a variety of cancer cell lines, according to the accumulating data from in vitro cancer models
- ✓ PI3K/Akt/mTOR, MAPK/ERK, and NF- κ B signaling are just a few of the carcinogenic pathways that EGCG routinely targets. At the same time, it promotes mitochondrial-mediated death by modulating the Bax/Bcl-2 balance and activating caspase
- ✓ The anticancer effects are concentration-dependent; in experimental systems, micromolar dosages successfully decrease angiogenic and metastatic indicators, induce apoptosis, and inhibit proliferation
- ✓ Crucially, EGCG's promise as a supplement to traditional cancer treatments is highlighted by its low toxicity and capacity to increase chemosensitivity

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ACKNOWLEDGEMENT

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